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13.2. Alkanes:

Alkanes are saturated hydrocarbons represented by the general formula C_nH_{2n+2} where 'n' is the number of carbon atoms in the molecule. Methane CH_4, is the first member of alkane family. The successive members are ethane C_2H_6, propane C_3H_8,butane C_4H_{10}, pentane C_5H_{12} and so on. It is evident that each member differs from its proceeding or succeeding member by a $-CH_2$ group.

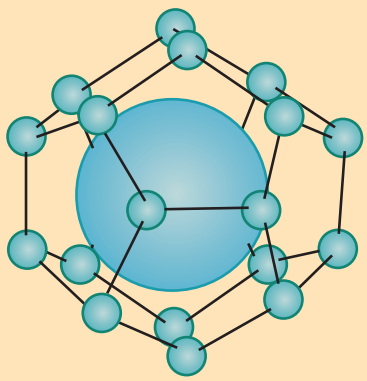
DO YOU KNOW?

Methane is the major component of the atmosphere of Jupiter, Saturn, Uranus and Neptune but only minor component of earth atmosphere. However, decomposition of plant and animal matter in an oxygen deficient environment like swamps, marshes, bogs and the sediments of lakes produces methane gas. So, it is otherwise known as marsh gas. It is also associated with coal and petroleum deposits.

DO YOU KNOW?

"FLAMMABLE ICE"

This catchy phrase describes a frozen mixture of water and methane gas chemically known as methane clathrates. The methane molecule which is produced by biological process under the deep-ocean, (at 4°C & 50 atm) doesn't simply reach the surface, instead each molecule is trapped inside clusters of 6 to18 water molecules forming methane clathrates. Many countries are working on how to tap out these vast resources of natural gas but mining and extracting are very difficult.



Nomenclature and isomerism:

We have already discussed the nomenclature of organic compound in Unit:11. Let us understand the nomenclature and isomerism in few examples. The first three members methane CH_4, ethane C_2H_6 and propane C_3H_8 have only one structure.